

Module 5 — Heavy Flavour & Quarkonia

Supplementary to the sPHENIX Standalone Course

1. Heavy flavour in Pythia8

Heavy quarks (c, b) are produced in three channels in Pythia's leading-order framework: direct pair production ($gg \rightarrow c\bar{c}$, $q\bar{q} \rightarrow c\bar{c}$), flavour excitation, and gluon splitting in the shower. Enable them via `HardQCD:hardccbar` and `HardQCD:hardbbbar`.

```
HardQCD:hardccbar = on      // gg→cc̄, qq̄→cc̄
HardQCD:hardbbbar = on      // gg→bb̄, qq̄→bb̄
PhaseSpace:pTHatMin = 5.0
```

2. Open vs hidden heavy flavour

	Examples	Flag family
n	D, D^*, Λ_c	HardQCD:hardccbar + decays
rm	$J/\psi, \chi_c, \psi(2S)$	Charmonium:all
m	B, B_s, Λ_b	HardQCD:hardbbbar + decays
om	$Y(1S,2S,3S), \chi_b$	Bottomonium:all

3. Quarkonium production: the NRQCD picture

Pythia implements quarkonium production in Non-Relativistic QCD (NRQCD): the cross section is a sum over colour-singlet and colour-octet matrix elements (LDMEs) times short-distance partonic processes. The LDMEs are tuneable.

	PDG ID	Typical pT reach (RHIC)
	443	0 – 10 GeV
	100443	0 – 8 GeV
	10441	feed-down to J/ψ
	20443	feed-down to J/ψ
	445	feed-down to J/ψ
	553	0 – 15 GeV
	100553	0 – 12 GeV
	200553	0 – 10 GeV

4. Feed-down chains

Most measured J/ψ are 'prompt' but many come from χ_c radiative decays and $\psi(2S)$ feed-down. For inclusive J/ψ analyses you need all sources:

```
Charmonium:all = on
# This enables:
#   gg -> J/psi g, gg -> J/psi gg, ...
#   gg -> chi_c0 g, gg -> chi_c1 g, gg -> chi_c2 g
#   gg -> psi(2S) g, ...
```

5. $B \rightarrow J/\psi X$

A fraction of J/ψ come from B-hadron decays ('non-prompt'). To include them, run open-bottom and let the B decays propagate:

```
HardQCD:hardbbbar = on  
PhaseSpace:pTHatMin = 5.0  
# Branching ratios for B->J/psi X are already in ParticleData
```

Note: Non-prompt J/ψ have a decay length (B mean $c\tau \approx 450 \mu\text{m}$); prompt are from the primary vertex. Experimentally the separation uses a secondary vertex or pseudo-proper decay time.

6. LHE files for matched samples

For NLO accuracy you can generate matrix elements externally (MG5_aMC, POWHEG) to a Les Houches Event (LHE) file, then feed it into Pythia for showering + hadronization + decays.

```
Beams:frameType = 4                // read LHE
Beams:LHEF = hard_events.lhe
PartonShowers:model = 1            // simple shower
SpaceShower:pTmaxMatch = 1         // emit up to pT_hat
```

7. Analysis example: $J/\psi \rightarrow \mu\mu$

```
TH1F h("mll", "m_{ll}", 120, 2.4, 3.6);
while (pythia.next()) {
    std::vector<Pythia8::Particle> mus;
    for (int i=0; i<pythia.event.size(); ++i) {
        auto& p = pythia.event[i];
        if (std::abs(p.id())==13 && p.isFinal()) mus.push_back(p);
    }
    for (size_t a=0; a<mus.size(); ++a)
        for (size_t b=a+1; b<mus.size(); ++b)
            if (mus[a].id() * mus[b].id() < 0) {
                auto s = mus[a].p() + mus[b].p();
                h.Fill(s.mCalc());
            }
}
```

8. Cross sections at RHIC (FYI)

Process	$\sigma \times \text{BR}$ at $\sqrt{s} = 200 \text{ GeV}$
$J/\psi \rightarrow e^+e^-$ (inclusive)	$\sim 8 \text{ nb}$
$\psi(2S) \rightarrow e^+e^-$	$\sim 1 \text{ nb}$
$Y(1S) \rightarrow e^+e^-$	$\sim 9 \text{ pb}$
$Y(2S)+Y(3S) \rightarrow e^+e^-$	$\sim 4 \text{ pb combined}$

These numbers depend on the PDF set and tune; treat them as sanity-check order-of-magnitude only.

9. Troubleshooting

Symptom	Likely cause
Zero events after <code>CharmTHatMin</code> higher than J/ψ mass	<code>Charmonium:all</code> turned on only singlet
No J/ψ in final state	Decays not turned on: check <code>443:onMode</code>
All J/ψ are prompt — no feed down	<code>Charmonium:all</code> turned on only singlet
Non-prompt fraction too high	B decays overshadow if <code>pTHatMin</code> is too high for charm